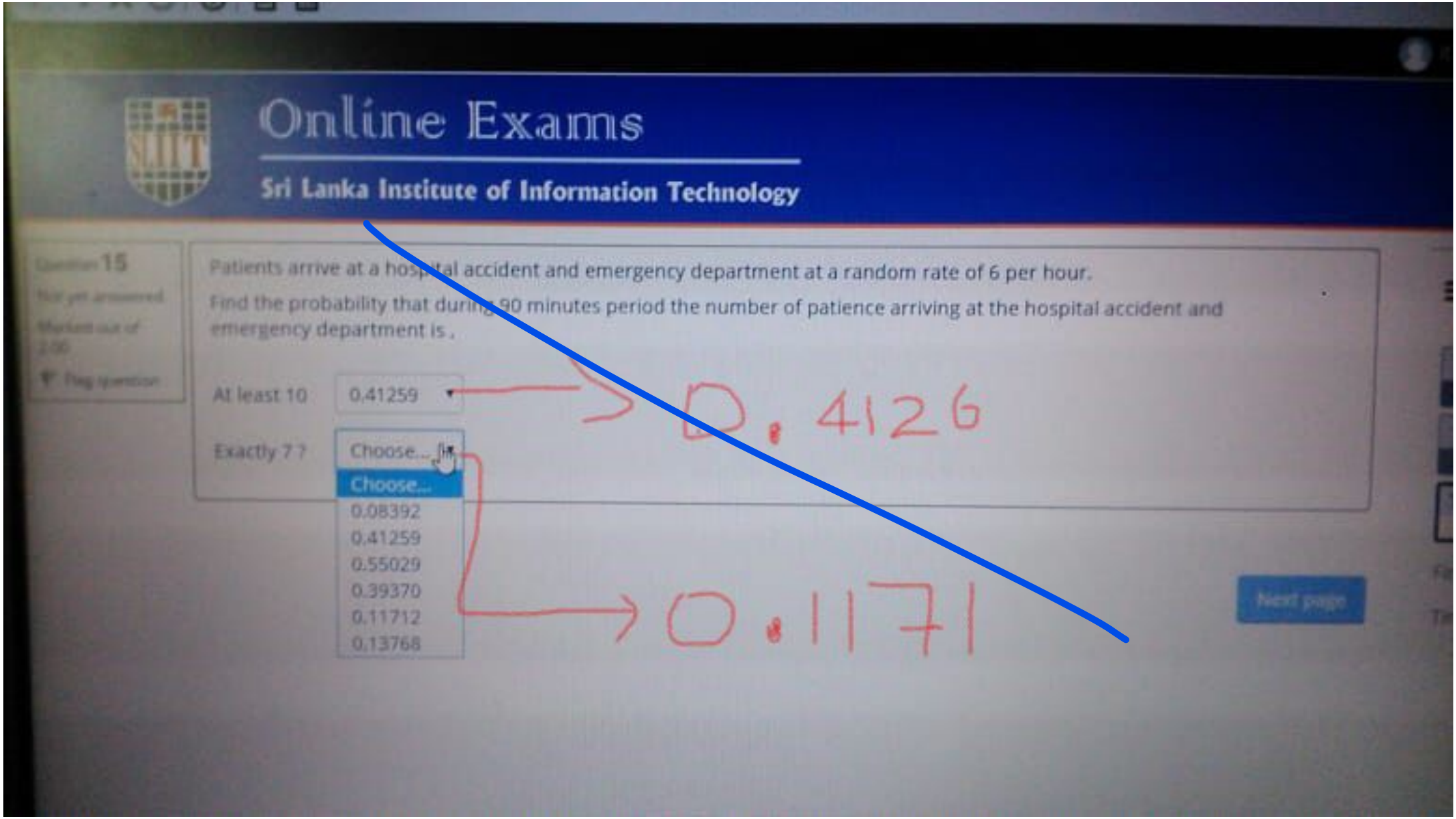
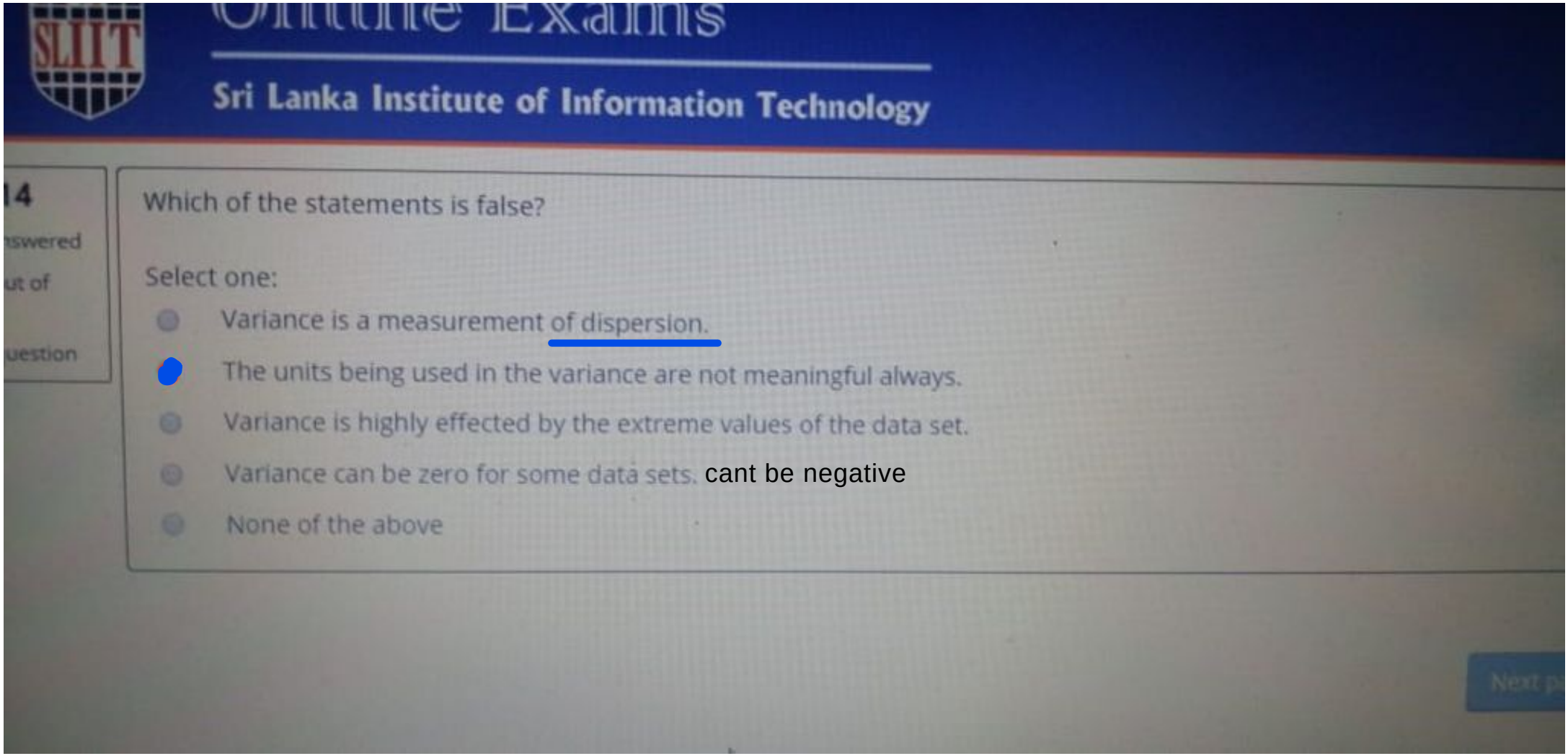

Online Exams
Sri Lanka Institute of Information Technology
Question 15
Not yet answered
Marked out of 2.00
Flag question
Patients arrive at a hospital accident and emergency department at a random rate of 6 per hour.
Find the probability that during 90 minutes period the number of patience arriving at the hospital accident and emergency department is ,
At least 10
0.41259
Exactly 7 ?
Choose...
Choose...
0.08392
0.41259
0.55029
0.39370
0.11712
0.13768
0.4126
0.1171
Next page


Online Exams
Sri Lanka Institute of Information Technology
14
swered
ut of
uestion
Which of the statements is false?
Select one:
Variance is a measurement of dispersion.
The units being used in the variance are not meaningful always.
Variance is highly effected by the extreme values of the data set.
Variance can be zero for some data sets.
cant be negative
None of the above

Online Exams

Sri Lanka Institute of Information Technology

In an experiment of rolling two dice, the first die shows a ONE and the other die rolls under the table and you cannot see it. Now, what is the probability that both die show ONE?

Select one:

- 1/3
- 1/6
- 1/36
- 9/36
- None of the above

Next page

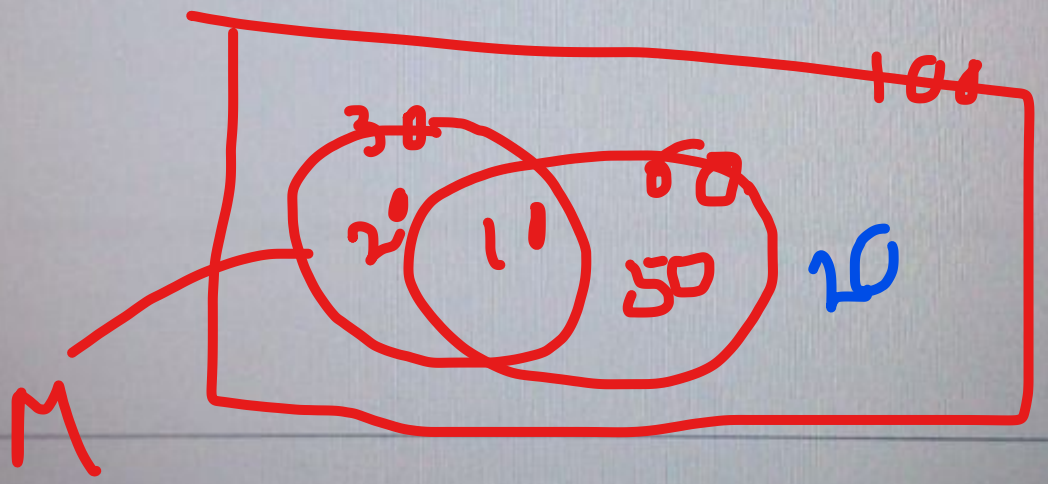
Online Exams

Sri Lanka Institute of Information Technology

In a classroom, 30% of students are doing Mathematics while 60% are doing Statistics. 10% are doing both of these subjects. What percentage is doing none of these subjects?

Select one:

- 0.6
- 0.3
- 0.5
- 0.2
- 0.1



Next page

Online Exams

Sri Lanka Institute of Information Technology

Suppose that in a national survey of 500 randomly selected adults, each person is asked how important religion is to him or her (very, fairly, not very), and whether the person favors or opposes stricter regulation of what can be broadcast on network television.

How many variables are there? Choose...

What is the sample size?

Choose...
550
500
Not given
600
3
4
2

2 ✓

500 ✓

Next page

Online Exams

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The collection and summarization of the socioeconomic and physical characteristics of the employees of a particular firm is an example of

Select one:

- Inferential statistics.
- Descriptive statistics.
- A parameter.
- A statistic.
- None of the above.

Next page

Online Exams

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A man sells electronic devices in his shop at an average rate of **2 per day**. What is the probability that the man sells at least 3 "electronic devices" in a **week (7days)** ?

lamda = 2*7 = 14

Select one:

- 0.99991
- 0.65412
- 0.85456
- 0.21454
- 0.75442

P(x>=3) = 0.99991

vered

of

stion

Following is **not** an example for a survey error.

Select one:

- Sampling error
- Measurement error
- Selection bias
- Non-response bias
- None of the above

Online Exams

Sri Lanka Institute of Information Technology

Question 15

Not yet answered

Marked out of 2.00

Flag question

Patients arrive at a hospital accident and emergency department at a random rate of 6 per hour.

Find the probability that during 90 minutes period the number of patience arriving at the hospital accident and emergency department is ,

lamda = 6 / 60 * 90 = 9

At least 10 — 0.41259

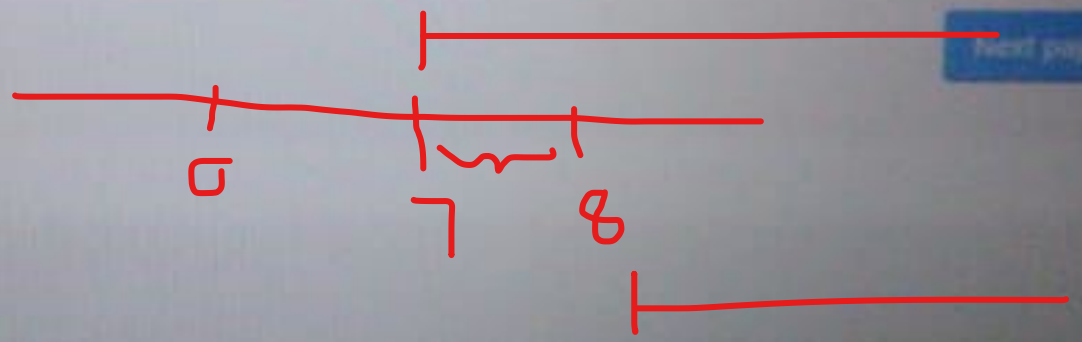
P(x>=10) = 0.41259

Exactly 7 ? — Choose...

- Choose...
- 0.08392
- 0.41259
- 0.55029
- 0.39370
- 0.11712
- 0.13768

P(x=7) = P(x>=7) - P(x>=8)
= 0.79322 - 0.67610
= 0.11712

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13

The mean and the median of the set given below

11, 26, 43, 15, 60, 18, 25

Select one:

- Mean = 28.29 and Median = 25
- Mean = 25 and Median = 28.5
- Mean = 27.29 and Median = 26.5
- Mean = 28.5 and Median = 27
- None of the above

Online Exams

Sri Lanka Institute of Information Technology

Find mean and standard deviation of the following data set.

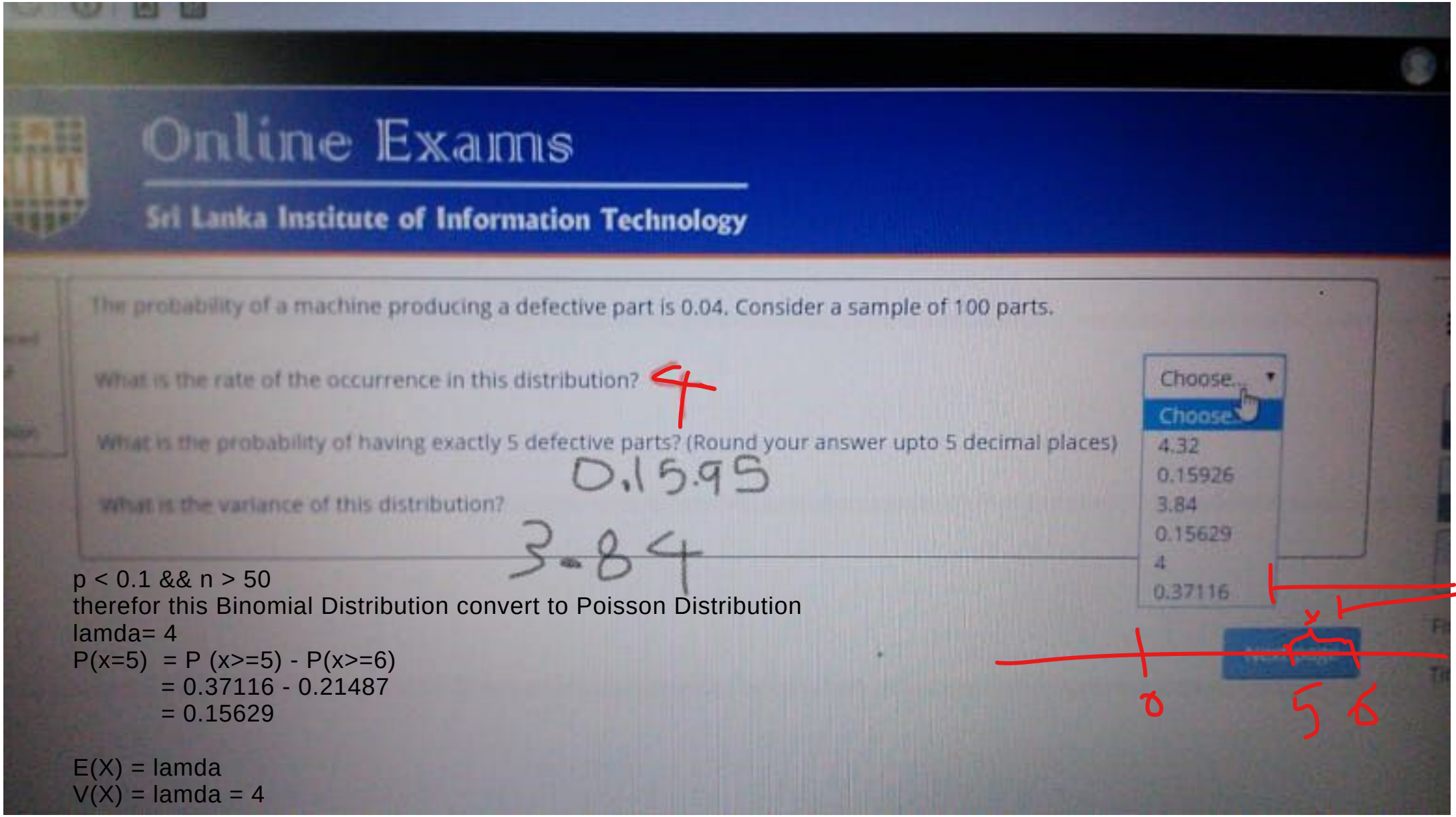
13.5, 12.6, 14.6, 19.2, 11.6, 24.3, 17.7

16.214

Mean Choose... ▾ 16.214

4.154

Standard deviation Choose... ▾ 4.154



p < 0.1 && n > 50
therefor this Binomial Distribution convert to Poisson Distribution
lamda= 4
P(x=5) = P (x>=5) - P(x>=6)
= 0.37116 - 0.21487
= 0.15629

E(X) = lamda
V(X) = lamda = 4

Online Exams

Sri Lanka Institute of Information Technology

Let 'X' be the number of passengers in a particular vehicle and it's probability distribution. Find the probability of having at least two passengers in a given vehicle ?

x	0	1	2	3	4
P(X=x)	0.2	0.1	a	0.3	0.1

Select one:

- 0.5
- 0.7
- 0.6
- 0.2
- 0.4

$0.2+0.1+a+0.3+0.1 = 1$

$a = 0.3$

$0.3 + 0.3 + 0.1 = 0.7$

Online Exams

Sri Lanka Institute of Information Technology

Question 5

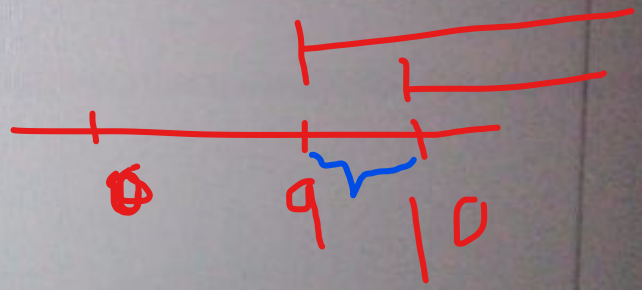
Not yet answered

Marked out of 1.00

Flag question

In a Car Sale an average of 3 out of every 5 customers would make a business. A random sample of 10 were selected. Find the probability that at least 9 people would make a business.

Binomial



Select one:

- 0.2546
- 0.0146
- 0.0245
- 0.2145
- 0.0464

n = 10

P(X) = 3/5 = 0.6

P(x=9) = P(x=9) - P(x=10)
= 10C9 * 0.6^9 * (1-0.6)^(10-9) + 10C10 * 0.6^10 * (1 - 0.6)^(10-10)
= 0.0464

Online Exams

Sri Lanka Institute of Information Technology

13

The mean and the median of the set given below

11, 26, 43, 15, 60, 18, 25

Select one:

- Mean = 28.29 and Median = 25
- Mean = 25 and Median = 28.5
- Mean = 27.29 and Median = 26.5
- Mean = 28.5 and Median = 27
- None of the above

Sri Lanka Institute of Information Technology

When selecting individuals for the sample, some individuals in the population have no chance of being selected to the sample (Or some individuals are excluded from the sampling frame). This survey error is known as,

Select one:

- Sampling error
- Measurement error
- Coverage error
- Non-response bias
- None of the above

Next page

≡ Quiz

1 2 8 9 15 16

Finish atte

Time left

Online Exams

Sri Lanka Institute of Information Technology

Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be 0 job arrivals.

Select one:

- 0.28811
- 0.41136
- 0.17492
- 0.03655
- 0.13534

Next page

Sri Lanka Institute of Information Technology

X	2	3	4	5
P(X=x)	0.2	0.3	a	0.3

Find E(2X+5)

$0.2 + 0.3 + a + 0.3 = 1$

$a = 0.2$

Select one:

- 15.2
- 11.2
- 12.2
- 12.5
- 10.5

$E(2x+5) = 2E(x) + E(5)$

$= 2*(0.2*2 + 0.3*3 + 0.2*4 + 0.3 * 5) + 5$

$=12.2$

Online Exams

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stion 1

yet answered

ked out of

Flag question

In a class room, there are 50 students. You are asked to analyze the relationship between Age and Height of students. What is the **suitable graphical tool** which can be used to describe these two variables together ?

Select one:

- Pie Chart
- Scatter Plot
- Bar Chart
- Clustered Bar Chart
- Histogram

Next page

Quiz navig

1 2 3

8 9 10

15 16 17

Finish attempt ...

Time left 0:59:40

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Question 18

Not yet answered

Marked out of 1.00

Flag question

X	-1	0	1	2
P(X=x)	0.3	0.2	a	0.3

Find $P(0 \leq X \leq 2)$?

Select one:

- 0.7
- 0.9
- 0.8
- 0.3
- 0.6

$0.3+0.2+a+0.3 = 1$

$a = 0.2$

$P = 0.2 + 0.3 + 0.2$

$= 0.7$

Next page

Quiz navigati

1 2 3 4

8 9 10 11

15 16 17 18

Finish attempt ...

Time left 0:01:33

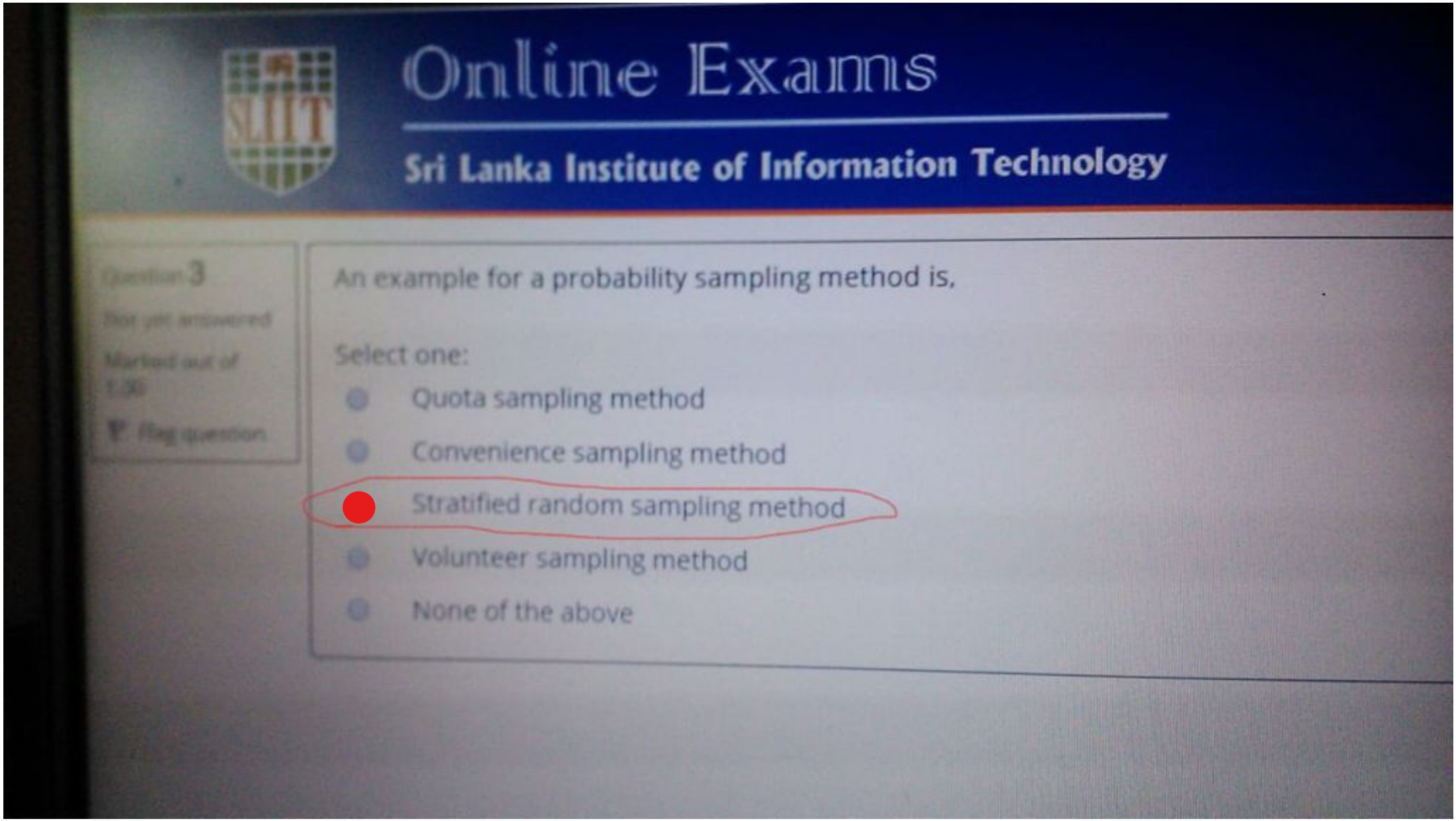
Sri Lanka Institute of Information Technology

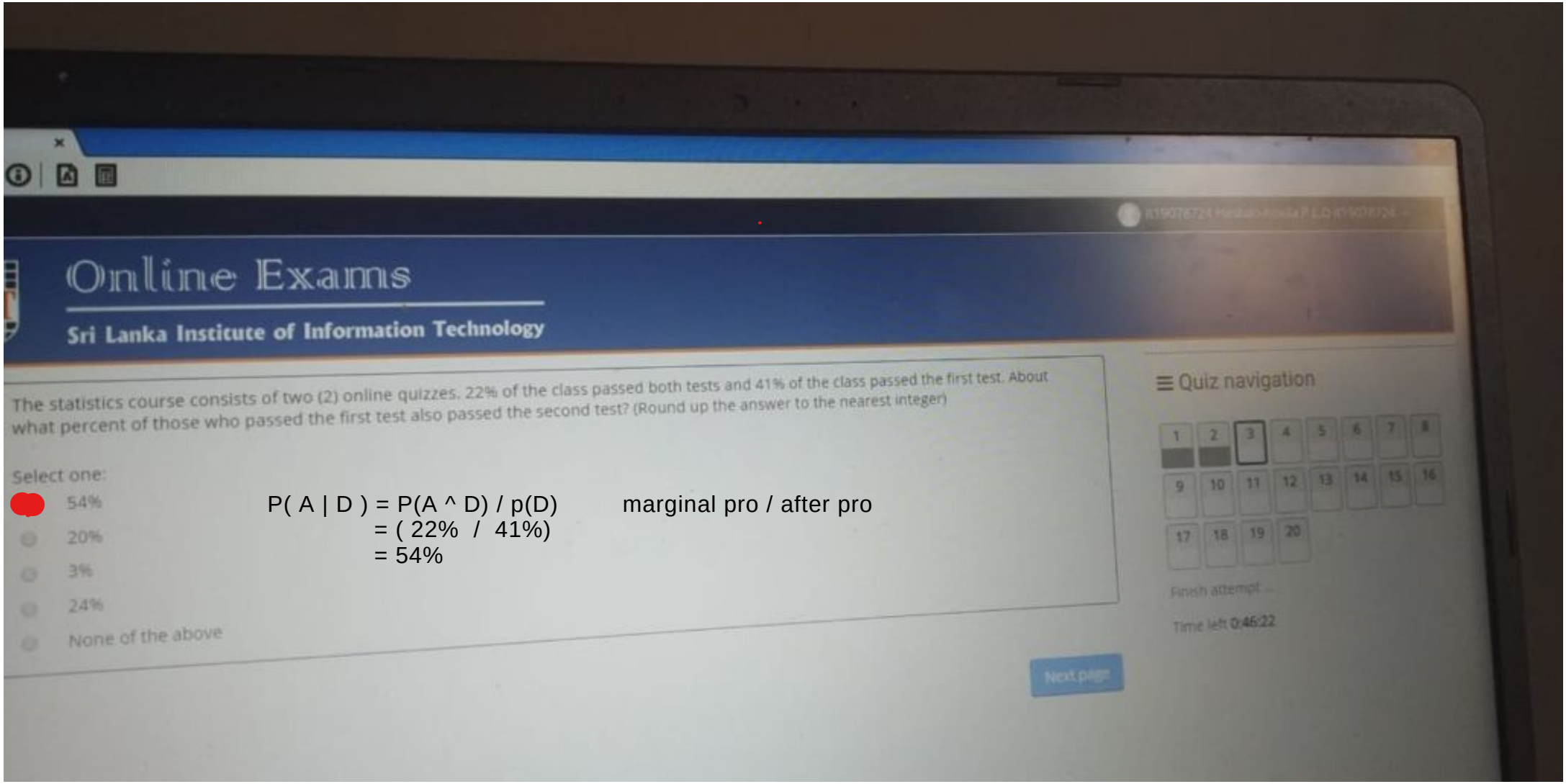
Select the **incorrect** statement?

Select one:

- Histogram can be used to describe numerical variables
- Pie chart can be used to describe one categorical variable donate pie
- Two way table can be used to describe two categorical variables
- Boxplot can be used to describe only one categorical variable
- Stem and leaf plot can be used to describe numerical variables.

One categorical & one numerical variable:
Parallel box plots


Online Exams
Sri Lanka Institute of Information Technology
Question 3
Not yet answered
Marked out of 1.00
Flag question
An example for a probability sampling method is.
Select one:
Quota sampling method
Convenience sampling method
Stratified random sampling method
Volunteer sampling method
None of the above


Online Exams
Sri Lanka Institute of Information Technology
The statistics course consists of two (2) online quizzes. 22% of the class passed both tests and 41% of the class passed the first test. About what percent of those who passed the first test also passed the second test? (Round up the answer to the nearest integer)
Select one:
54%
20%
3%
24%
None of the above
P(A | D) = P(A ^ D) / p(D)
= (22% / 41%)
= 54%
marginal pro / after pro
Quiz navigation
1 2 3 4 5 6 7 8
9 10 11 12 13 14 15 16
17 18 19 20
Finish attempt ...
Time left 0:46:22
Next page

Online Exams

Sri Lanka Institute of Information Technology

Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be 0 job arrivals.

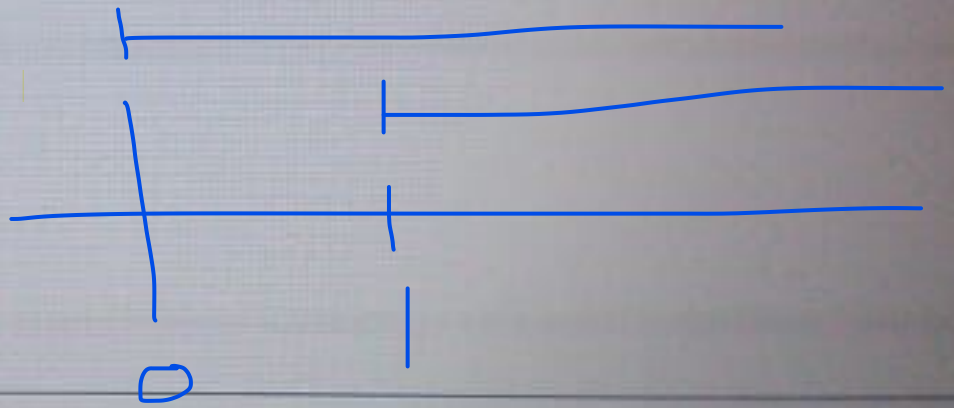
Select one:

- 0.28811
- 0.41136
- 0.17492
- 0.03655
- 0.13534

lamda = 2

per - Poisson Distribution

P(x=0) = P(x>=0) - P(x>=1)
= 1.00000 - 0.86466
= 0.132334



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Online Exams

Sri Lanka Institute of Information Technology

An example for a nominal categorical variable is.

Select one:

- Importance of culture to respondent (very, somewhat, or not very important)
- Opinion about a new political law (favor or oppose)
- Length of forearm from elbow to wrist (in centimeters)
- Number of songs on a digital music player
- None of the above

Moodle

Online Exams

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Question 1

Not yet answered

Marked out of 1.00

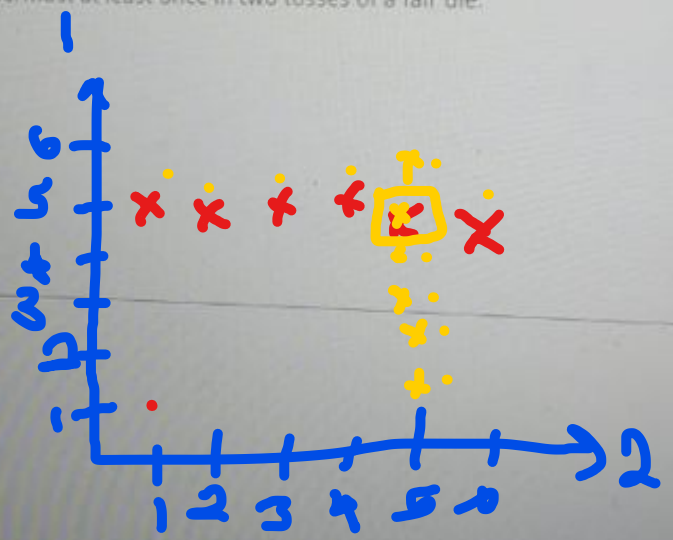
Flag question

Find the probability of a 5 turning uppermost at least once in two tosses of a fair die.

Select one:

- 11/36
- 12/36
- 15/36
- 14/36
- None of the above

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Online Exams

Sri Lanka Institute of Information Technology

Question 1

Not yet answered

Marked out of 1.00

Flag question

Which of the following is **not** a form of non-probability sampling?

Select one:

- Quota sampling.
- Convenience sampling.
- Cluster sampling.
- Purposive/judgement sampling.
- They are all forms of non-probability sampling.